
Reinforcement Learning, Assignment 4

Rwik Kamilya (s4822285)

Abstract

We implement Proximal Policy Optimization with clipping (PPO-Clip) from scratch on CartPole-v1 and study which design choices are responsible for its stability and sample efficiency. PPO extends basic Advantage Actor-Critic by reusing on-policy rollouts for multiple minibatch updates while constraining policy drift through a clipped probability-ratio objective. We evaluate default PPO, four ablations, and a lightly tuned PPO configuration under a 10^6 environment-step budget. Default PPO solves CartPole in all seeds but is slower than expected, reaching the 475 return threshold after 356762 ± 129743 steps. The tuned configuration substantially improves sample efficiency, reaching the threshold after 154317 ± 27985 steps and achieving final evaluation return 500.0 ± 0.0 . Ablations show that repeated update epochs are essential, while clipping, GAE and advantage normalization affect stability and early learning. Previous DQN, REINFORCE, AC and A2C results are included as contextual baselines.

1. Introduction

The previous assignments followed a standard progression in deep reinforcement learning. Tabular methods first exposed the core ideas of value estimation, exploration and backup targets. DQN then replaced the table by a neural approximation of $Q(s, a)$, making value-based learning applicable to continuous observations but introducing instability from bootstrapping and shared parameters (Mnih et al., 2015). REINFORCE, AC and A2C moved in a different direction: instead of acting greedily with respect to a learned value function, they directly parameterized the policy and used a critic or advantage estimate to reduce gradient variance (Williams, 1992; Sutton & Barto, 2018).

However, a basic actor-critic update remains fragile. If a policy-gradient step is too large, the new policy may differ substantially from the behaviour policy that generated the sampled data. This is especially problematic when one tries to improve sample efficiency by reusing the same on-policy rollout for several gradient steps. PPO addresses this

problem by retaining the actor-critic structure but replacing the plain advantage-weighted policy loss with a clipped surrogate objective (Schulman et al., 2017). The clipping term limits the incentive to move the new policy too far from the old policy, giving a simple first-order approximation to trust-region behaviour.

This report studies PPO-Clip on CartPole-v1. The central question is: *which PPO mechanisms matter most for sample efficiency and robustness: clipping, generalized advantage estimation, advantage normalization, or repeated minibatch updates?* We answer this by implementing PPO from scratch, ablating the main engineering choices, and then evaluating a tuned variant suggested by the ablation results. The aim is not merely to show that PPO can solve CartPole, but to identify which parts of the method explain its learning behaviour.

2. Theory

We consider an episodic Markov decision process with states s_t , actions a_t , rewards r_t , discount γ , and stochastic policy $\pi_\theta(a_t|s_t)$. PPO is an actor-critic method: the actor represents the policy and the critic estimates a state-value function $V_\theta(s_t)$. In our implementation both heads share a two-layer neural body, and the actor defines a categorical distribution over the two CartPole actions.

The starting point is the advantage actor-critic objective. Given an advantage estimate A_t , a basic policy-gradient update increases the log-probability of actions with positive advantage and decreases it for actions with negative advantage. PPO modifies this update by comparing the current policy to the old policy that collected the rollout. Define the probability ratio

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} = \exp(\log \pi_\theta(a_t|s_t) - \log \pi_{\theta_{\text{old}}}(a_t|s_t)). \quad (1)$$

The clipped PPO objective is

$$L^{\text{clip}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)], \quad (2)$$

where ϵ is the clipping coefficient. The first term is the usual importance-weighted policy-gradient surrogate. The second term clips the ratio to the interval $[1 - \epsilon, 1 + \epsilon]$. Thus, once the new policy has moved sufficiently far from the old policy

in a direction that would artificially improve the surrogate objective, the objective is flattened. We minimize $-L^{\text{clip}}$ in the code. The no-clip ablation replaces this objective by the unclipped loss $-\mathbb{E}_t[r_t(\theta)A_t]$.

The critic is trained by regression to return targets R_t :

$$L^V(\theta) = \frac{1}{2} \mathbb{E}_t[(R_t - V_\theta(s_t))^2]. \quad (3)$$

The complete minimized objective is

$$L(\theta) = -L^{\text{clip}}(\theta) + c_v L^V(\theta) - c_e \mathbb{E}_t[\mathcal{H}(\pi_\theta(\cdot|s_t))], \quad (4)$$

where c_v controls critic weight and c_e controls entropy regularization. The entropy term discourages premature collapse to a deterministic policy.

Advantages are computed with generalized advantage estimation (GAE) (Schulman et al., 2016). With temporal-difference residual

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad (5)$$

GAE forms

$$A_t^{\text{GAE}(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}. \quad (6)$$

The parameter λ interpolates between low-variance one-step TD estimates and higher-variance long-horizon estimates. In the $\lambda = 0$ ablation, PPO uses only one-step residual information.

Relative to A2C, PPO makes three important changes. First, the policy loss is clipped through $r_t(\theta)$ rather than using the plain advantage loss. Second, the same rollout is reused over multiple shuffled minibatch epochs. Third, this reuse is stabilized by practical choices such as GAE, advantage normalization, entropy regularization and gradient clipping.

Algorithm 1 Implemented PPO-Clip

- 1: Initialize shared actor-critic network and rollout buffer.
 - 2: Collect T on-policy transitions; store rewards, values and old log-probabilities.
 - 3: Bootstrap through truncations and compute GAE advantages backwards.
 - 4: Set return targets $R_t = A_t + V(s_t)$.
 - 5: For K epochs, shuffle the rollout into minibatches.
 - 6: Recompute log-probabilities, values and entropy for stored actions.
 - 7: Minimize clipped actor loss + value loss – entropy bonus.
 - 8: Clip gradients, log KL/clip fraction, and evaluate periodically.
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3. Experiments

3.1. Setup

All experiments use CartPole-v1 (Towers et al., 2024). The observation has four continuous components and the action space contains two discrete actions. Each non-terminal step gives reward +1, and the maximum episode return is 500. We use deterministic evaluation with greedy actions every 10,000 environment steps over 20 episodes.

The default network is a shared two-layer MLP with 64 hidden units, Tanh activations, orthogonal initialization, a categorical actor head and a scalar critic head. Default PPO uses $\gamma = 0.99$, $\lambda = 0.95$, learning rate $3 \cdot 10^{-4}$, rollout length 2048, minibatch size 256, 10 update epochs, clip coefficient $\epsilon = 0.2$, entropy coefficient 0.01, value-loss coefficient 0.5, and gradient clipping at norm 0.5. Training runs for 10^6 environment steps.

We evaluate default PPO, a tuned PPO variant, and four ablations. The tuned configuration uses shorter rollouts (512), smaller minibatches (64), fewer update epochs (4), lower entropy coefficient (0.001), and disables advantage normalization. The ablations isolate individual mechanisms: no clipping, $\lambda = 0$, no advantage normalization, and a single update epoch. PPO variants are summarized by area under the learning curve over the first 200k steps (AULC_{200k}), first solve time t_{475} , solve rate, and final evaluation return. A run is considered solved once its evaluation return reaches 475.

The implementation explicitly uses clipped policy ratios, minibatch SGD, multiple epochs over each rollout, GAE, advantage normalization unless ablated, entropy regularization, gradient clipping, approximate-KL logging, clip-fraction logging, seed aggregation, and correct Gymnasium termination/truncation handling. It does not use SAC, replay buffers, target networks, reward normalization, observation normalization, vectorized environments, learning-rate schedules, target-KL early stopping, model checkpointing, or explained-variance logging.

For context, we compare against saved Assignment 2/3 results for DQN, REINFORCE, AC and A2C. This comparison is useful for scale but not perfectly protocol-matched: PPO uses deterministic evaluation checkpoints, while previous baselines are reconstructed from saved training-return traces.

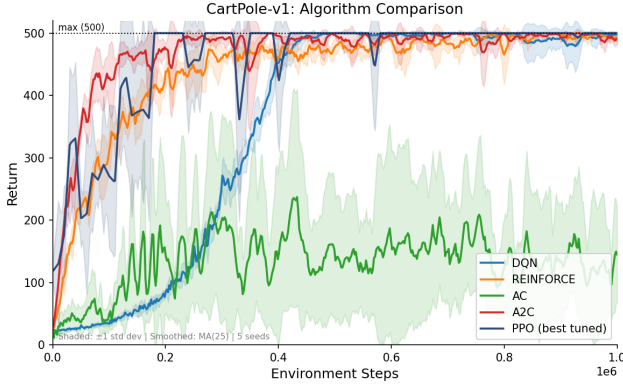
3.2. Results

Table 1 shows that default PPO solves CartPole in all seeds, but not especially quickly: it reaches the 475 threshold after 356762 ± 129743 steps. The tuned PPO configuration is much stronger, reducing solve time to 154317 ± 27985 steps and reaching final evaluation return 500.0 ± 0.0 . Thus, PPO

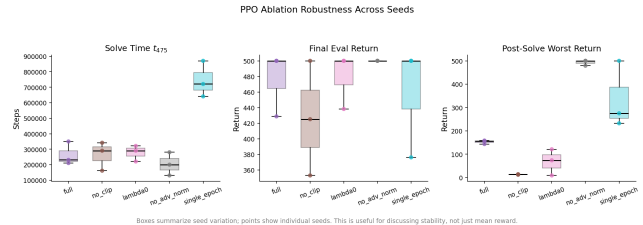
Table 1: Summary metrics from the final study. PPO variants are the main controlled comparison. DQN, REINFORCE, AC and A2C are contextual previous-assignment baselines. Higher $AULC_{200k}$ is better; lower t_{475} is better.

Method	$AULC_{200k}$	t_{475} steps	Solve rate	Evaluation note
PPO full	0.373	356762 ± 129743	100%	final return 485.7 ± 28.5
PPO tuned	0.587	154317 ± 27985	100%	final return 500.0 ± 0.0
PPO no clip	0.268	264875 ± 75828	100%	ablation
PPO $\lambda = 0$	0.389	277845 ± 41983	100%	ablation
PPO no adv. norm	0.493	204117 ± 61082	100%	ablation
PPO single epoch	0.193	744107 ± 95070	100%	ablation
A2C [†]	0.797	105629 ± 17153	100%	previous assignment
REINFORCE [†]	0.623	266654 ± 24617	100%	previous assignment
AC [†]	0.186	412956 ± 183552	40%	previous assignment
DQN [†]	0.103	415389 ± 17630	100%	previous assignment

[†]Contextual baselines: saved training-return traces, not deterministic PPO-style evaluation logs.



(a) Learning curves against previous baselines.



(b) Seed-level PPO robustness diagnostics.

Figure 1: PPO learning and robustness. The cross-assignment comparison is contextual; the PPO variants form the controlled experiment.

is highly sensitive to implementation and hyperparameter choices even in CartPole.

The ablations clarify which choices matter. The single-epoch variant is the weakest PPO configuration, with $AULC_{200k} = 0.193$ and $t_{475} = 744107 \pm 95070$. This indicates that repeated minibatch updates are central to PPO’s sample efficiency: without data reuse, the method behaves more like a slow on-policy actor-critic learner. Removing clipping also hurts early learning, giving $AULC_{200k} = 0.268$. Although this variant still solves eventually, the loss no longer restricts policy-ratio changes, so repeated updates are less controlled. The $\lambda = 0$ variant performs better than no clipping but remains slower than the tuned configuration, suggesting that one-step advantages are usable in CartPole but less effective than GAE for stable improvement.

The no-advantage-normalization ablation is especially informative. It achieves $AULC_{200k} = 0.493$ and $t_{475} = 204117 \pm 61082$, making it the strongest single ablation and motivating the tuned configuration. This result should not be read as a general rejection of advantage normalization. In dense-reward CartPole, raw advantage magnitudes may provide useful scale information, whereas in harder tasks normalization often improves conditioning.

Compared with previous baselines, A2C has the best early-learning $AULC_{200k} = 0.797$ and fastest t_{475} . PPO tuned nevertheless reaches perfect final evaluation return and is far stronger than default PPO. Because the previous baselines are not logged with the same deterministic evaluation protocol, we interpret this comparison as historical context rather than a strict ranking.

4. Discussion

The main lesson is that PPO’s empirical behaviour is determined by the interaction between its surrogate objective and its engineering choices. Clipping is theoretically motivated as a policy-drift control mechanism, but clipping alone does not guarantee fast learning. Repeated minibatch epochs provide the data efficiency that distinguishes PPO from a simpler on-policy actor-critic method. GAE supplies a more flexible advantage estimator than one-step TD, while entropy regularization and gradient clipping contribute additional numerical stability.

The strongest empirical result is the tuned PPO configuration. Its improvement over default PPO is substantial: $AULC_{200k}$ increases from 0.373 to 0.587, and t_{475} decreases from about 357k to 154k environment steps. This suggests that the default PPO settings were conservative or poorly matched to CartPole. Shorter rollouts and minibatches allow more frequent updates; fewer epochs reduce excessive reuse; lower entropy regularization permits sharper exploitation once a good policy emerges; and disabling advantage normalization preserves useful reward-scale information in this simple dense-reward setting.

There are several limitations. First, the comparison to DQN, REINFORCE, AC and A2C is not fully apples-to-apples, because those baselines are saved training traces rather than deterministic evaluation checkpoints. Second, CartPole is low-dimensional and dense-reward, so conclusions about advantage normalization or entropy strength may not transfer to sparse-reward or continuous-control environments. Third, the implementation omits optional PPO refinements such as learning-rate annealing, target-KL early stopping, explained-variance logging, vectorized environments and checkpointing. These omissions are acceptable for a from-scratch implementation, but they delimit the strength of the empirical conclusions.

5. Conclusion

We implemented PPO-Clip from scratch and evaluated it on CartPole-v1 through controlled ablations and a tuned final configuration. The algorithm satisfies the actor-critic structure of previous assignments while replacing the plain A2C policy update with a clipped probability-ratio objective. Default PPO solves CartPole reliably, but the ablation study shows that repeated rollout reuse, GAE, clipping and normalization choices strongly shape learning speed and stability. The tuned configuration solves substantially faster than default PPO and reaches perfect final evaluation return. Future work should rerun all previous baselines under the same deterministic evaluation protocol, extend the sweep to harder environments, and test whether the no-advantage-normalization finding persists beyond CartPole.

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